

CSE 347/652|DES306/525
Usable Security and Privacy

Course Project Guidelines

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Course Project

- *Final source code: due November 17, 2026 midnight,*
submitted through the project link in Classroom
 - At least 50% of the source code must be original
- *The executable needs to be posted on a file server by the deadline.* (We may be able to provide cloud simulators on our VMs, you have to inform us earlier).

Project Phases

- Phase I – User Studies to evolve Requirements, Design and Workflows
 - Deadline: October 4, 2026
 - *Deliverables: User Study Evaluation reports, Requirements, Design Prototypes*
- Phase II – Development of the Tool
 - Deadline: November 17, 2026
 - *Deliverables: Executable for Testing and User Guide*

Poster Presentations

- There will be two poster sessions
 - First poster presentation will focus on compilation of usability studies and design themes for your project
 - Will be held on *October 7, 2026*
 - The second Poster presentation will focus on demonstrations on usability and security features of your project. Will include Live Demonstrations
 - Will be held on *November 26, 2026*

Usability Studies

- Prior to creating prototypes for their project, teams must also engage in design research processes
- Processes must focus on usability aspects of the application, and collect relevant qualitative and quantitative data which will then fuel the development.
- Studies can be carried out through primary and secondary research methods, eg: questionnaires, focus groups, contextual interviews, comparative analyses of existing applications.

Course Project (Cont.)

- Testing of other groups' projects
 - The Executable and user-guide for each group will be posted on Classroom for your testing before 8 a.m., November 18, 2026
 - The testing must be completed and the results reported by individual students (not by groups).
 - *Each student will be assigned to test at least 8 group projects and must complete the assigned testing during the period Noon November 18, 2026 to midnight November 25, 2026.*

Course Project (Cont.)

- Submit your test report on each of the assigned group projects by *November 28, 2026 midnight*
- Your test report for each group project (not including your own) should include the following information:
 - Describe different features for the Project in terms of usability?
 - Find vulnerabilities in the features, you must show the validity on each vulnerability you report and include relevant screenshots.

Course Project (Cont.)

- Final presentation (through recorded video):
 - Presentation will be online (recorded) *no more than 20 minutes* and include the following:

Part A:

1. How did you design your usability studies?
2. How did your usability studies help you better design your tool?
3. Present your design themes and methodology

Part B:

1. What *security requirements* you have identified?
2. How have you addressed these security requirements?
3. A list of classes of all the *vulnerabilities* reported by students
4. Cause of vulnerabilities and lessons learnt

- Deadline: *Midnight November 29, 2026*

Evaluation

- Development (20%)
 - Usability of the tool (25%)
 - Functional correctness (10%)
 - Quality of developed software (15%)
 - Presentations (10%)
 - Final group presentation (15%)
 - Satisfaction of security requirements (25%)
 - Peer evaluation from group members (varying impact)
- Usability and Security Testing (15%)
 - Participation in Usability studies (5% - 1% for every five studies you participate)
 - Usability evaluation of course projects and security testing (10% - 0.5% for every functional/security bug)
- One best project will be given additional 5% bonus credit

Project Topics

- Password Managers
- Secure Storage Systems
- Authentication systems
- Calendar agent (speech)
- Permission Managers
- Any domain that you know well (need prior approval from TAs)